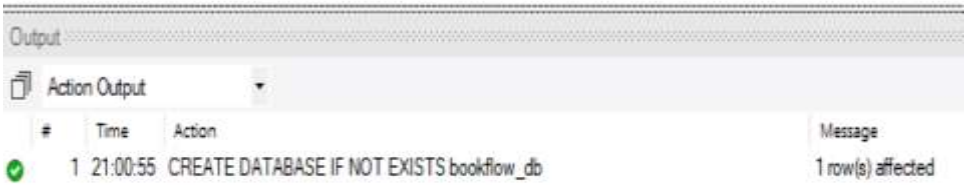
WEEK-2

K.Harini
2520030399

Step 1 — Create Database

```
1 •  CREATE DATABASE IF NOT EXISTS bookflow_db;
```



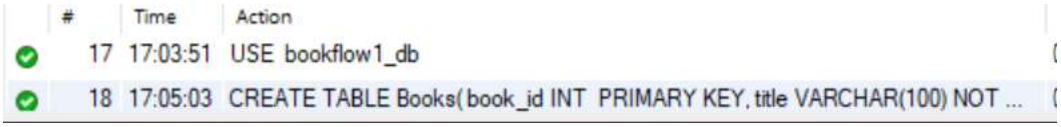
Step 2 — Select Database

```
1 •  CREATE DATABASE IF NOT EXISTS bookflow_db;
2 •  USE bookflow_db;
```



Step 3— Create Books Table (DDL + Constraints)

```
1 •  CREATE DATABASE IF NOT EXISTS bookflow1_db;
2 •  USE  bookflow1_db;
3 •  CREATE TABLE Books(
4    book_id INT  PRIMARY KEY,
5    title VARCHAR(100) NOT NULL,
6    isbn VARCHAR(20) UNIQUE,
7    published_year INT CHECK(published_year<2027)
8    );
```



Step 4- Display Books

```
9 •  SELECT * FROM Books;
```

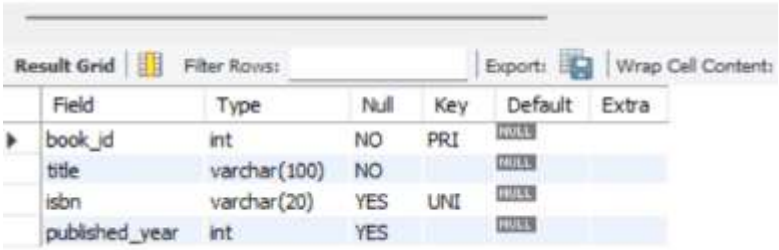


Result Grid | Filter Rows:

book_id	title	isbn	published_year
NULL	NULL	NULL	NULL

Step 5— Insert books and display

```
10 •  INSERT INTO books(title,isbn,published_year)VALUES
11    ('The Alchemist','9374679545265',1988),
12    ('Clean Code','859956356965',2008),
13    ('Atomic habits','3897503565036',2015);
14 •  DESCRIBE Books;
```

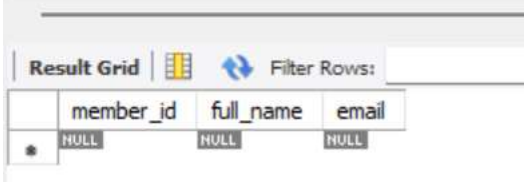


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
title	varchar(100)	NO		NULL	
isbn	varchar(20)	YES	UNI	NULL	
published_year	int	YES		NULL	

Step 6—Members table (create, insert, display)

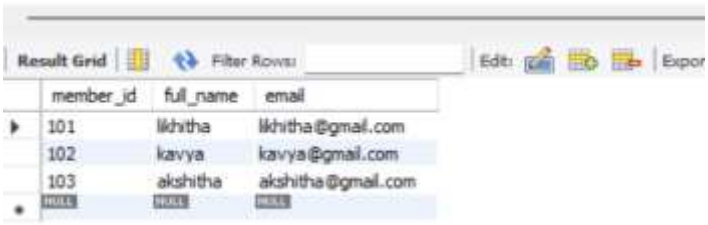
```
15 •  CREATE TABLE Members(
16    member_id INT PRIMARY KEY,
17    full_name VARCHAR(100),
18    email VARCHAR(150) UNIQUE
19    );
20 •  SELECT * FROM Members;
```



Result Grid | Filter Rows:

member_id	full_name	email
NULL	NULL	NULL

```
21 •  INSERT INTO Members(member_id,full_name,email)VALUES
22    ('101','likhitha','likhitha@gmail.com'),
23    ('102','kavya','kavya@gmail.com'),
24    ('103','akshitha','akshitha@gmail.com');
25 •  SELECT * FROM Members;
```



Result Grid | Filter Rows: | Edit: | Expor

member_id	full_name	email
101	likhitha	likhitha@gmail.com
102	kavya	kavya@gmail.com
103	akshitha	akshitha@gmail.com
NULL	NULL	NULL

Step 7— Create Loans table(FOREIGN KEYS)

```
26 • CREATE TABLE Loans(
27   loan_id INT PRIMARY KEY,
28   member_id INT,
29   book_id INT,
30   Loan_date DATE,
31   FOREIGN KEY (member_id) REFERENCES M
32   FOREIGN KEY (book_id) REFERENCES Boo
33   );
34 • SELECT * FROM Loans;
```



Step 8— Insert data into loans and display

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01
NULL	NULL	NULL	NULL

Step 7— GROUP BY Query(books per year)

```
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;
```

	published_year	Total_Books
▶	1925	1
	1949	1
	1960	1

Step 8 — Donation via TRANSACTION(atomicity)

```
CREATE TABLE Donation_History (
 donation_id INT PRIMARY KEY,
 book_id INT,
 donor_name VARCHAR(100),
 donation_date DATE,
 FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
START TRANSACTION;
INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);
INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());
COMMIT;

SELECT * FROM Books;
```

	book_id	title	isbn	published_year
▶	1	The Great Gatsby	9780743273565	1925
	2	To Kill a Mockingbird	9780061120084	1960
	3	1984	9780451524935	1949
	4	Animal Farm	9780451526342	1945
*	NULL	NULL	NULL	NULL

SELECT * FROM Donation_History;

	donation_id	book_id	donor_name	donation_date
▶	1	4	Raj Kumar	2026-08-10
	NULL	NULL	NULL	NULL

Step 9 – Index on ISBN + FAST SEARCH

```
CREATE INDEX idx_books_isbn ON Books(isbn);
SELECT * FROM Books WHERE isbn = '9780451524935';
```

	book_id	title	isbn	published_year
▶	3	1984	9780451524935	1949
*	NULL	NULL	NULL	NULL